

NMEC595 - Research Methodology

Lecture 3: Defining the Research Problem

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Lecture Outline

- 1 What is a Research Problem?
- 2 Selecting the Research Problem
- 3 Necessity of Defining the Problem
- 4 Techniques for Defining a Problem
- 5 Problem Definition in Practice
- 6 Types of Research Problems
- 7 Key Takeaways

Definition of a Research Problem

A **research problem** is a challenge or issue identified by a researcher that needs to be addressed. It may be:

- **Theoretical:** Concerning conceptual or theoretical matters
- **Practical:** Concerning actual situations

Key Criteria: A research problem exists if:

- 1 The problem is recognized by an individual, group, or organization
- 2 There are at least two available courses of action
- 3 There are at least two potential outcomes, with one being preferable
- 4 There is uncertainty about which action to select

Essential Components of a Research Problem

Five Essential Components:

- 1 **Individual/Group**: The person, group, or organization facing the problem
- 2 **Objective(s)**: The intended outcome to be achieved
- 3 **Alternative Means**: Availability of at least two possible courses of action
- 4 **Doubt/Uncertainty**: Uncertainty regarding which course of action is preferable
- 5 **Environment**: The context or setting in which the problem arises

Note: Addressing a research problem involves selecting the most appropriate solution to achieve the objective within a particular environment.

Guidelines for Problem Selection

When choosing a research problem, consider the following:

- **Avoid topics that have been extensively studied:** Limited opportunity to contribute new knowledge (not a definitive suggestion, depends on motivation, capability, and timeline)
- **Avoid highly controversial issues:** May present challenges for most researchers
- **Avoid problems that are too narrow or too vague:** Such problems are often impractical
- **Choose topics that are both familiar and feasible:** Ensure relevant sources are accessible

Criteria for Problem Selection

Key Selection Criteria:

- 1 **Importance:** The topic should be significant (subjective)
- 2 **Researcher's qualification:** The researcher should have relevant training and background
- 3 **Feasibility:** Access to necessary data, facility, and cooperation should be possible
- 4 **Cost:** The project should be achievable within the available budget
- 5 **Time:** The research should be completed within the allotted timeframe
- 6 **Preliminary study:** Consider if a feasibility study is required

Critical: The selected problem should engage the researcher and be given high priority.

Sources for Problem Ideas

How to Find Research Ideas:

- **Consult experts or professors:** Obtain advice from experienced researchers (with a good heart!)
- **Review recent literature:** Examine new articles and publications on relevant topics
- **Consider existing techniques:** Explore how established methods may address other problems
- **Engage in discussions:** Exchange ideas and seek constructive feedback
- **Draw from practical experience:** Recognize issues arising from professional practice

Remember: Research problems should originate from the researcher's own inquiry, not from external borrowing.

Why Define the Problem?

“A problem clearly stated is a problem half solved”

Defining a research problem is essential because it:

- **Discriminates data:** Identifies which data are relevant and which are not
- **Keeps the researcher focused:** Maintains direction throughout the research process
- **Guides planning:** Supports effective organization and planning of the study
- **Determines data collection:** Clarifies which data need to be gathered
- **Guides methodology:** Helps select suitable techniques and methods for analysis

Consequences of Poor Problem Definition

Impact of Ill-Defined Problems:

- Researcher cannot plan strategy effectively
- Cannot discriminate relevant from irrelevant data
- Creates hurdles in research execution
- Leads to collection of unnecessary data
- Results in wasted time, effort, and resources
- May prevent meaningful conclusions

Bottom Line: A well-defined problem is prerequisite for successful research.

Two-Step Process for Problem Formulation

Problem Formulation involves:

① Understanding the problem thoroughly

- Discuss with colleagues and subject experts
- Seek guidance from experienced researchers/supervisors
- Review available literature and background materials

② Rephrasing into analytical/operational terms

- Express problem in specific, measurable terms
- Make problem scientifically testable
- Define pertinent terms clearly

Understanding the Problem

How to Understand the Problem:

- **Discuss with colleagues:** Share perspectives and get feedback
- **Consult research guide/supervisor:** Benefit from experience and guidance
- **Examine available literature:**
 - Conceptual literature (theories and concepts)
 - Empirical literature (related research studies)
- **Identify available resources:** Determine what materials and data are accessible

This review provides foundation for precise problem specification.

Rephrasing into Analytical Terms

Making Problem Specific:

- **Use operational definitions:** Define terms in measurable, observable ways
- **Specify relationships:** Clarify what variables or factors are involved
- **Set boundaries:** Define scope and limitations
- **Make it testable:** Problem must be subject to investigation
- **Use sequential approach:** Each formulation more specific than the previous

Example: General problem “Component failure occurs frequently in mechanical systems”

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Example: General problem “Component failure occurs frequently in mechanical systems” → Specific problem “What factors contribute to fatigue failure in steel shafts used in automotive transmissions?”

Sequential Pattern in Problem Formulation

Iterative Process:

- First formulation: Broad, general statement of problem
- Intermediate formulations: Progressively narrower and more specific
- Final formulation: Precise, analytical, realistic terms
- Each successive formulation:
 - More specific than preceding one
 - Phrased in more analytical terms
 - Realistic in terms of available data and resources

Note: Multiple revisions are normal and expected in problem formulation.

Problem Definition in Different Contexts

How Problem Selection Differs by Setting:

● Academic institutions:

- Guide or advisor introduces the problem in broad terms
- Researcher refines the problem and defines it in operational terms
- *Example: A professor suggests exploring new materials for heat sinks; the student focuses on optimizing fin design for thermal performance*

● Business/Government:

- Administrative agency or management identifies the problem
- Researcher examines the background and context of the problem
- Emphasis is placed on practical needs and effective solutions
- *Example: NVIDIA's R&D team is assigned the problem of improving GPU cooling efficiency; engineers analyze thermal issues and develop new cooling technologies for their latest graphics cards*

In every context, the researcher must develop a thorough understanding of the problem and define it with precision.

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 - Example: Study “effect of lubrication type on wear in ball bearings” instead of “study of wear in machines.”
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- **Discriminate data:** The definition should specify which data are relevant and which are not.
 - Example: Only collect data related to lubricant properties and wear rate, not unrelated machine characteristics.

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 - Example: “Does increasing coolant flow rate reduce tool wear?” is testable.
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 - Example: Define what is meant by “wear,” “efficiency,” or “fatigue limit” in the context of the study.

Characteristics of Well-Defined Problems (3/3)

A Good Problem Definition Should:

- **Consider feasibility:** The problem should be realistic and achievable given available resources (time, funding, equipment, expertise).
 - Example: Attempting a large-scale experimental study may not be feasible if only limited laboratory equipment is available.

My take: *The problem should be rich enough to generate more research problems.*

Summary: A well-defined problem provides clarity, focus, and direction, ensuring that research efforts are effective and efficient.

Key Takeaways

- **The first step** in research is to identify and define the problem
- A problem should include **five essential elements**: individual or group, objectives, alternative options, uncertainty, and context
- **Selection requires attention** to feasibility, significance, and researcher's background
- **Formulation involves two stages**: first, thorough understanding; second, clear operational rephrasing
- **Defining the problem** is the most critical stage of the research process
- **Use an iterative process**: Refine the problem statement through repeated review for greater specificity
- **A clear problem definition** supports effective planning and execution of research

Summary: Steps in Problem Definition

1 Understand thoroughly

- Discuss with experts
- Review literature (conceptual and empirical)

2 Examine available resources

- Identify accessible materials and data
- Assess feasibility

3 Rephrase into operational terms

- Make specific and measurable
- Define key terms

4 Refine iteratively

- Each formulation more precise than previous
- Sequential narrowing from general to specific

Questions and Discussion

Key Question to Ask

“Have I defined my research problem precisely enough that another researcher could replicate and understand my study?”

Thank you!